A Survey on Inference Optimization Techniques for Mixture of Experts Models

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The emergence of large-scale Mixture of Experts (MoE) models represents a significant advancement in artificial intelligence, offering larger model capacity and computational efficiency through conditional computation. However, deploying and running inference on these models presents significant challenges in computational resources, latency, and energy efficiency. This comprehensive survey analyzes optimization techniques for MoE models across the entire system stack. We first establish a taxonomical framework that categorizes optimization approaches into model-level, system-level, and hardware-level optimizations. At the model level, we examine architectural innovations including efficient expert design, attention mechanisms, various compression techniques such as pruning, quantization, and knowledge distillation, as well as algorithm improvement including dynamic routing strategies and expert merging methods. At the system level, we investigate distributed computing approaches, load balancing mechanisms, and efficient scheduling algorithms that enable scalable deployment. Furthermore, we delve into hardware-specific optimizations and co-design strategies that maximize throughput and energy efficiency. This survey provides both a structured overview of existing solutions and identifies key challenges and promising research directions in MoE inference optimization. To facilitate ongoing updates and the sharing of cutting-edge advances in MoE inference optimization research, we have established a repository accessible at https://github.com/MoE-Inf/awesome-moe-inference/.

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1 Introduction

Large language models (**LLMs**) have revolutionized artificial intelligence, demonstrating unprecedented capabilities across various domains including natural language processing [21, 131, 187], computer vision [34, 37, 230], and multimodal tasks [94, 141, 193]. Models like GPT-4 [130], Claude [8], and Gemini [179] have achieved remarkable performance in tasks ranging from natural language understanding to complex reasoning and code generation. The impressive capabilities of these models are largely attributed to their massive scale, both in terms of model parameters and computational resources invested in training. This scaling trend is supported by empirical evidence showing consistent improvements in model performance with increased size, as demonstrated by various scaling laws in language modeling and other domains [6, 20, 79]. However, this trajectory presents significant challenges in terms of computational efficiency and resource utilization, particularly during inference, where real-world deployment constraints become critical [10, 204, 222, 239].

Mixture of Experts (**MoE**) has emerged as a promising architectural solution to address scaling challenges in machine learning [160], which was originally introduced by Jacobs et al. in the early 1990s as a method for learning subtasks in neural networks [73]. Numerous MoE-based models [41, 57, 185] have been developed over the years. In the era of LLMs, MoE has again experienced a renaissance [1, 30, 75, 175]. The core principle of MoE is to distribute the model's parameters across multiple specialized sub-networks, or experts, with a learned routing mechanism that selectively activates only the relevant experts for each input. This approach allows models to maintain a large parameter count while keeping computational costs manageable through sparse activation. Recent implementations, such as Mixtral 8x7B [75], DeepSeek-V3 [33], and DBRX [183], have demonstrated the effectiveness of this strategy in scaling language models to trillions of parameters while maintaining reasonable computational requirements.

The success of MoE in scaling models has led to its adoption in various state-of-the-art systems. For example, Google's GLaM [39] outperforms GPT-3 while using significantly fewer computational resources during inference. Similarly, Mixtral 8x7B [75] has demonstrated competitive performance compared to much larger dense models, while maintaining efficient inference characteristics. DeepSeek-V3 [33], a recent open-source MoE model, has surpassed other open-source alternatives and demonstrated performance comparable to prominent closed-source models such as GPT4-o and Claude-3.5-Sonnet. Table 1 summarizes recent state-of-the-art open-source MoE models that have garnered significant attention, showcasing their rapid evolution and widespread adoption across major tech companies and research institutions. The models range in size from 6.7B to 1043B parameters, with varying architectures characterized by different numbers of experts, hidden layers, and attention heads. The chronological progression from 2022 to late 2025 demonstrates increasing model sizes and architectural sophistication, with recent models like DeepSeek-V3 pushing the boundaries in terms of parameter count and model performance. This trend further highlights the strong potential of the MoE architecture in advancing the field of LLMs. These successes have sparked widespread interest in MoE across both academia and industry, leading

Table 1. A List of SoTA MoEs

Reference	Para.	Experts	#L	#H	d_{model}	d_{ffn}	d_{expert}	Affiliation	Time
NLLB [25]	54B	2/64/0	24	16	1024	8192	8192	FaceBook	2022.07
Qwen2-57B-A14B [209]	57.4B	8/64/0	28	28	3584	18944	2560	Alibaba	2023.05
Mixtral-8x7B [75]	46.7B	2/8/0	32	32	4096	14336	14336	Mistral AI	2023.12
OpenMoE [206]	34B	2/16/0	12	12	768	2048	2048	NUS et al.	2023.12
DeepSeekMoE [30]	16.4B	6/64/2	28	16	2048	10944	1408	DeepSeek-AI	2024.01
Qwen1.5-MoE [182]	14.3B	4/60/0	24	16	2048	5632	1408	Alibaba	2024.02
JetMoE [162]	8.52B	2/8/0	24	32	2048	5632	5632	MIT et al.	2024.03
Jamba [104]	51.6B	2/16/0	32	32	4096	14336	14336	ai21labs	2024.03
DBRX [183]	132B	4/16/0	40	48	6144	10752	10752	Databricks	2024.03
Grok-1 [200]	314B	2/8/0	64	48	6144	UNK	UNK	xAI	2024.03
Arctic [150]	482B	2/128/0	35	56	7168	4864	4864	Snowflake	2024.04
Mixtral-8x22B [75]	141B	2/8/0	56	48	6144	16384	16384	Mistral AI	2024.04
DeepSeek-V2 [32]	236B	6/160/2	60	128	5120	12288	1536	DeepSeek-AI	2024.04
Skywork-MoE [197]	13B	2/16/0	52	36	4608	12288	12288	Kunlun Tech	2024.05
Yuan2 [198]	40B	2/32/0	24	16	2048	8192	8192	IEIT-Yuan	2024.05
LLaMA-MoE [241]	6.7B	2/8/0	32	32	4096	11008	11008	Zhu et al.	2024.06
OLMoE [122]	6.92B	8/64/0	16	16	2048	1024	1024	AllenAI	2024.07
Phi-3 [1]	41.9B	2/16/0	32	32	4096	6400	6400	MicroSoft	2024.08
GRIN-MoE [108]	41.9B	2/16/0	32	32	4096	6400	6400	MicroSoft	2024.09
Hunyuan-Large [175]	389B	1/16/1	64	80	6400	18304	18304	Tencent	2024.11
DeepSeek-V3 [33]	671B	8/256/1	61	128	7168	18432	2048	DeepSeek-AI	2024.12
MiniMax-Text-01 [121]	456B	2/32/0	80	64	6144	9216	9216	MiniMax-AI	2025.1
DeepSeek-R1 [31]	671B	8/256/1	61	128	7168	18432	2048	DeepSeek-AI	2025.1
Llama 4 Maverick [119]	402B	1/128/0	48	40	5120	16384	8192	Meta	2025.4
Qwen3-235B-A22B [208]	235B	8/128/0	94	64	4096	12288	1536	Alibaba	2025.5
ERNIE-4.5 [11]	300B	8/64/0	54	64	8192	28672	3584	Baidu	2025.6
Hunyuan-A13B [184]	80B	8/64/1	32	32	4096	24576	3072	Tencent	2025.6
Kimi-K2 [85]	1,043B	8/384/1	61	64	7168	18432	2048	MoonshotAI	2025.7
GPT-oss [3]	120B	4/128/0	36	64	2880	11520	2880	OpenAI	2025.8
GLM-4.5 [224]	355B	8/160/1	92	96	5120	12288	1536	Z.ai	2025.8
LongCat [181]	560B	12/512/0	28	64	6144	12288	2048	Meituan	2025.9

Param. represents the number of total parameters. Experts are listed according to the format of the number of activation experts, total experts, and shared experts. #L represents the number of hidden layers, #H represents the number of attention heads. d_{model} is the hidden size, d_{ffn} is the intermediate size of FFNs, d_{expert} is the intermediate size of FFN experts.

to innovations in model design [23, 195, 228], training techniques [38, 51, 110], and deployment strategies [16, 17, 217].

However, the efficient deployment of MoE models for inference presents unique and significant challenges [70, 177, 215, 234]. The dynamic nature of expert activation patterns (the computation path is determined by the input, see Section 2 for details) introduces complexity in resource management and scheduling, that is, not present in traditional dense models. At the model level, the design of efficient expert architectures faces challenges in balancing model capacity with computational efficiency, while routing mechanisms must optimize expert selection and load distribution. The system-level challenges are particularly complex: distributed computation requires sophisticated scheduling algorithms to manage expert placement and activation, load balancing must handle dynamic workload variations across experts, and memory management needs to efficiently handle the loading and unloading of expert parameters. Hardware-level challenges stem from the fundamental mismatch between traditional hardware architectures optimized for dense computation and

the sparse, dynamic nature of MoE inference. This necessitates specialized acceleration techniques to handle sparse computation patterns, manage memory bandwidth efficiently, and provide flexible computation capabilities for dynamic expert switching. Communication overhead in distributed settings presents another significant challenge, particularly when experts are distributed across different devices or nodes, requiring careful optimization of data movement and synchronization.

Numerous methods have been developed to address these challenges in MoE deployment and inference [77, 144, 156, 202]. While the rapid growth of research in this field demonstrates its importance, it can also make it difficult to identify key trends and best practices. A critical gap in the existing literature is the absence of a systematic framework for analyzing and developing comprehensive inference optimization solutions for MoE models. To bridge this gap, our work offers a comprehensive survey of inference optimization techniques for MoE models. We propose a taxonomical framework that categorizes optimization approaches into model-level, system-level, and hardware-level optimizations, as illustrated in Figure 1. This framework provides a structured approach to understanding and comparing different optimization techniques. While there are related surveys on LLM efficiency [10, 92, 98, 186, 189, 204, 222, 239] and MoE architectures [14, 45, 188], our work is the first to specifically focus on inference optimization techniques for MoE models. We systematically analyze optimization approaches at different abstraction levels, from model architecture to hardware acceleration, providing a valuable resource for researchers and practitioners working deploy MoE models for different real-world applications.

The remainder of this survey is organized as follows: Section 2 provides background on MoE models and their inference characteristics. Sections 3–5 detail optimization techniques at the model, system, and hardware levels, respectively. Section 6 discusses future challenges and opportunities, and Section 7 concludes the survey.

2 Fundamentals of Mixture of Experts

MoE represents a significant architectural paradigm in neural networks, particularly in LLMs, where it enables conditional computation through sparse activation mechanisms [14]. At its core, each layer of an MoE architecture consists of a routing network $R(x)$ and a set of expert networks $E_1, E_2, ..., E_N$, where N denotes the total number of experts. The fundamental principle of MoE block can be expressed as $y = \sum_{i=1}^{N} g_i(x) \cdot E_i(x)$, where $g_i(x)$ represents the gating function for expert i, and $E_i(x)$ is the output of expert i.

As illustrated in Figure 2, existing studies typically use the MoE module to replace part of the traditional dense layer, thereby forming a sparse MoE layer. While most research focuses on substituting the FFN module with the MoE module [1, 32, 75, 76, 182], some have also explored replacing the attention module [77, 162, 163, 228].

The typical MoE model inference procedure follows a sequence of expert selection, parallel computation, and output aggregation. First, the router computes expert selection probabilities:

$$\theta = \text{Softmax}(R(x)), \tag{1}$$

where $x \in \mathbb{R}^d$ is the input token embedding, $R(\cdot)$ is the routing function, and $\theta \in \mathbb{R}^N$ represents the expert selection probabilities for N total experts. Next, the top-k experts are selected based on these probabilities:

$$E_{\text{selected}} = \text{TopK}(\theta, K), \tag{2}$$

where E_{selected} contains the indices of the K selected experts, $K \leq N$. The selected experts then process the input in parallel:

$$y_i = E_i(x), \quad \forall i \in E_{\text{selected}}, \tag{3}$$

where $E_i(\cdot)$ represents the computation of expert i, and y_i is its output.

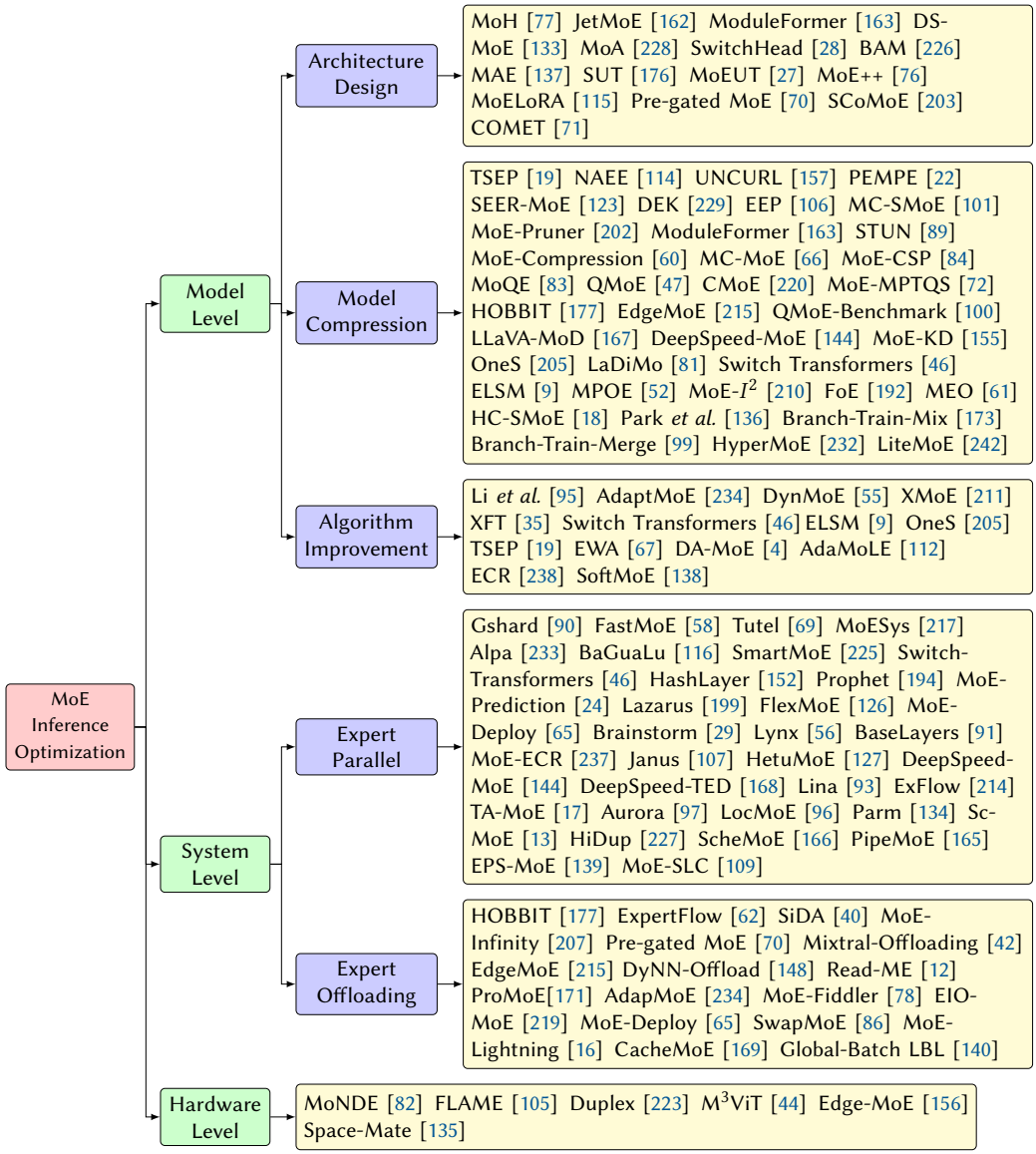


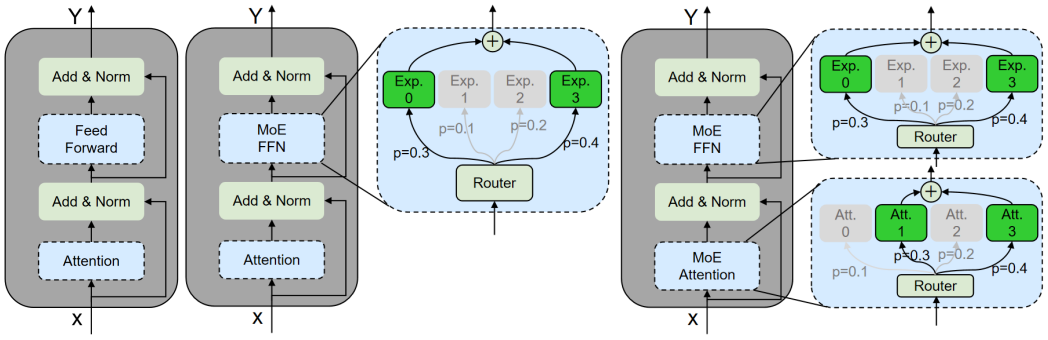
Fig. 1. Taxonomy of MoE inference optimization.

Finally, the expert outputs are combined through a weighted aggregation. In many MoE implementations, the routing weights, θ_i, are normalized to form a probability distribution over the selected experts. This is illustrated in Equation (4):

$$y = \sum_{i \in E_{\text{selected}}} \frac{\theta_i}{\sum_{j \in E_{\text{selected}}} \theta_j} \cdot y_i, \tag{4}$$

where the weights are normalized by their sum.

However, it is important to note that not all MoE approaches normalize these weights. For example, Switch-Transformer [46] uses a softmax to determine expert choice without afterwards



(a) Dense Layer (b) MoE FFN Layer (c) MoE Attention+FFN Layer

Fig. 2. Architectural comparison of dense layer with MoE layers: (a) conventional dense transformer layer, (b) transformer layer with MoE-based feed-forward network, and (c) transformer layer with both MoE-based attention and feed-forward networks.

normalization, simplifying the weighting scheme. In this case, the expert outputs is:

$$y = \sum_{i \in E_{\text{selected}}} \theta_i \cdot y_i. \tag{5}$$

More explicitly, some models provide a choice in this regard. The Qwen2-MoE [209] model, for example, includes a parameter named "norm_topk_prob" that specifically controls whether the routing weights for the top-k selected experts are normalized. This highlights that the normalization of expert weights is a key design choice in MoE architectures, with different models adopting different strategies.

The MoE architecture offers several compelling advantages that have contributed to its growing adoption in modern neural networks. First, conditional computation yields substantial efficiency gains. By activating only a subset of task-relevant experts for each input token, MoE models reduce computational cost while sustaining performance comparable to dense models with equivalent capacity. This property is particularly valuable in resource-constrained settings and when scaling to larger model sizes.

Second, expert specialization enhances modeling fidelity. Individual experts acquire domain- or pattern-specific competence, enabling more precise and targeted computations. This effect is especially beneficial in language modeling, where different experts can focus on distinct linguistic phenomena, domains, or tasks.

Third, the dynamic routing mechanism enables adaptive computation based on input complexity. More challenging or nuanced inputs can engage multiple experts with complementary specializations, while simpler inputs might require only a small subset of experts. This adaptive resource allocation helps optimize the computation-performance trade-off and potentially improves both efficiency and effectiveness.

By partitioning dense models into relatively independent expert models and dynamically activating specific subsets (or the entire set) of experts based on each input token, the model's overall performance can be significantly enhanced with only a marginal increase in inference computation. This approach clearly demonstrates the MoE model's exceptional flexibility and scalability.

3 Model-level Optimizations

Model-level optimizations aim to enhance the inherent structure and efficiency of MoE models through systematic improvements in architecture, parameter optimization, and algorithmic design.

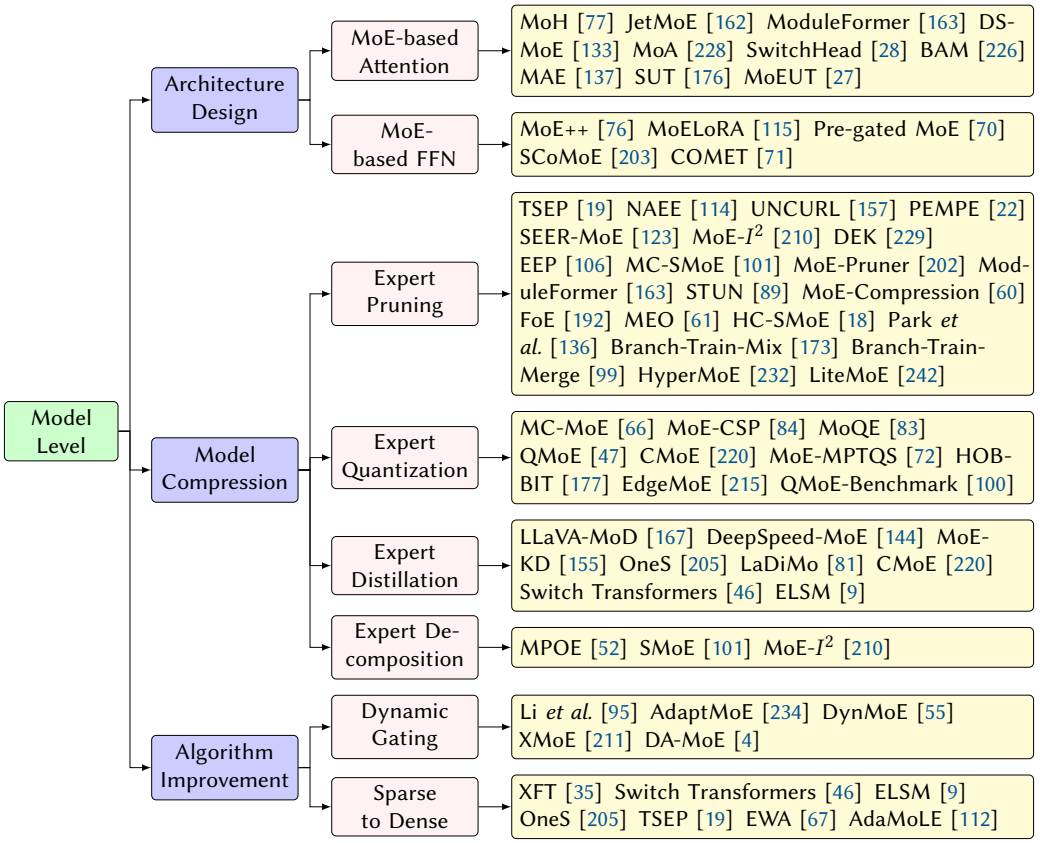


Fig. 3. Model-level inference optimization techniques for MoE.

These optimizations can be broadly categorized into three main areas: efficient model architecture design, model compression techniques, and algorithmic improvements. Architecture design focuses on developing more efficient expert and attention structures, while compression techniques aim to reduce model size and memory footprint through methods such as pruning, quantization, and knowledge distillation. Algorithmic improvements concentrate on enhancing the dynamic aspects of MoE models, including routing mechanisms and expert combination strategies. Figure 3 illustrates the detailed taxonomy of model-level optimization, that is, described in this section.

3.1 Efficient Model Architecture Design

A transformer block typically consists of two main components: the attention module and the FFN module. To build better MoE models, many studies focus on designing improved versions of both the attention and FFN modules, aiming to achieve strong performance while maintaining high efficiency.

3.1.1 MoE-based Attention Design. In addition to the typical application of the MoE structure in the FFN module of the transformer layer, current studies explore how to incorporate MoE into the attention module for improved performance. MAE [137] was the first to interpret multi-head attention as an MoE system, using a learned gating function to activate different experts—each composed of $n-1$ heads, where n denotes the number of attention heads per layer. However, its fixed expert composition limits flexibility in head selection. To address this, MoA [228] and BAM [226]

enable dynamic selection of k individual heads per token and share key/value projections across heads, significantly reducing memory overhead while preserving expressivity. SwitchHead [28] further improves efficiency by sharing query and key projections, though at the cost of reduced routing diversity. MoH [77] introduces shared heads and a two-stage routing mechanism, achieving better performance than MoA by balancing specialization and redundancy. Building on these ideas, ModuleFormer [163] extends sparsity to both attention and FFN layers, enabling modular model scaling. This design is adopted in JetMoE-8B [162], a high-performing open-source model with fully sparse layers. In contrast, DS-MoE [133] proposes a hybrid dense-training/sparse-inference paradigm, which eases optimization during training while retaining inference efficiency. Meanwhile, SUT [176] and MoEUT [27] achieve extreme parameter sharing across layers via sparse universal transformers, trading off some capacity for compactness and faster convergence.

3.1.2 MoE-based FFN Design. To enhance the efficiency of MoE-based models, current research explores various variants of the standard MoE module. MoE++ [76] introduces zero-computation experts that skip actual computation for certain tokens, effectively reducing FLOPs without degrading performance—particularly useful in latency-sensitive scenarios. SCoMoE [203] tackles the communication bottleneck in distributed MoE by adopting a hierarchical all-to-all communication pattern, which scales better on large clusters but requires careful topology-aware scheduling. Pregated MoE [70] improves inference latency on memory-constrained devices by prefetching experts based on early gating signals, though it assumes predictable routing patterns. COMET [71] replaces the standard linear router with a tree-based selector, enabling more fine-grained expert assignment at the cost of increased routing complexity. Additionally, MoELoRA [115] reimagines LoRA as a lightweight MoE, enabling parameter-efficient fine-tuning via routed adapters and offering a compelling trade-off between adaptability and storage overhead. MoE-Prism [201] offers a novel method to deconstruct monolithic experts into fine-grained sub-experts, which can enable more flexible and efficient MoE service delivery.

3.2 Model Compression Techniques

Model compression is a popular area of research for reducing model size in current LLM studies, with techniques such as pruning [102, 118], quantization [48, 191], knowledge distillation [2, 54], and low-rank decomposition [196, 221]. Since experts constitute the majority of the weights in MoE models (e.g., 96% for Mixtral-8x7B [75]), most MoE-related compression efforts focus on applying these common techniques specifically to the experts. Notably, pruning and quantization in MoE architectures must preserve expert diversity; overly aggressive compression can collapse expert specialization and degrade performance, a challenge less pronounced in dense models.

3.2.1 Expert Pruning. Due to the large number of parameters in MoE-based models, current research explores pruning methods to reduce the number of parameters in MoE experts. These methods are generally divided into structured and unstructured pruning. Most studies focus on structured expert pruning, which aims to reduce the number of experts in the MoE layer while maintaining model accuracy. As shown in Figure 4(a), some approaches directly remove unimportant experts (left side), while others merge groups of experts into a single expert (right side). Representative methods are summarized in Table 2. TSEP [19] removes non-professional experts for the target downstream task while fine-tuning professional experts to preserve model performance. They also demonstrate the superiority of the eager expert pruning paradigm over other possible solutions like two-pass optimization or staged expert pruning. NAEE [114] eliminates unimportant experts by evaluating expert combinations on a small calibration dataset to minimize accuracy loss, simultaneously reducing model sizes and increase inference speed, while UNCURL [157] reduces the number of experts based on MoE router logits. Some insights useful for model design choices

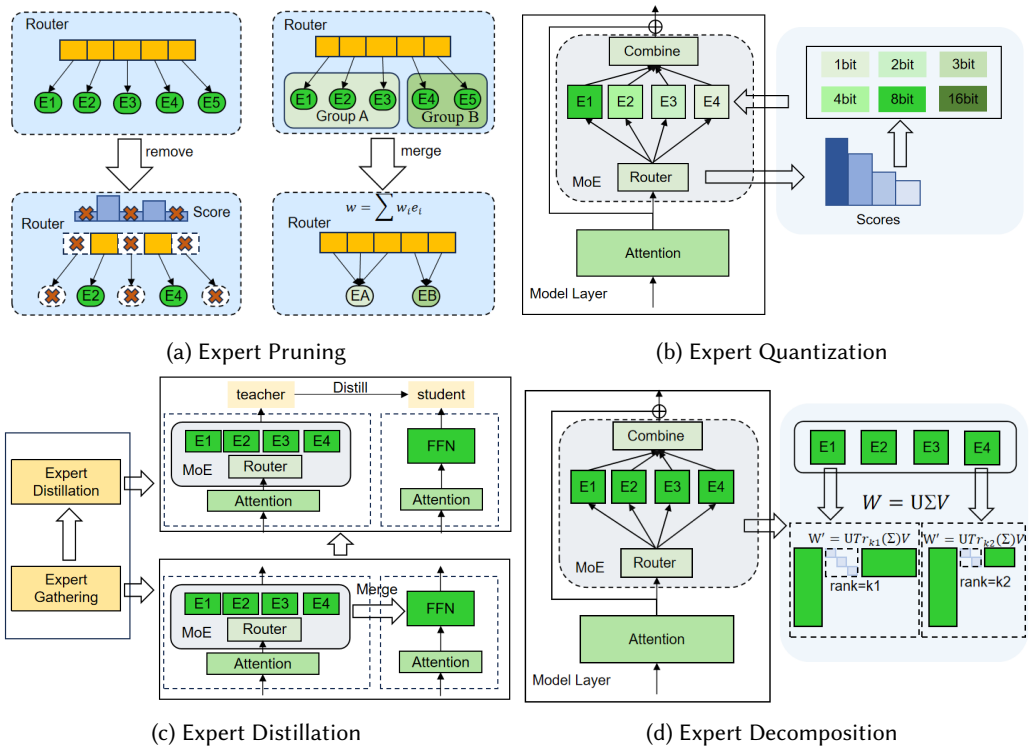


Fig. 4. Compression techniques for MoE models.

Table 2. A comparison of Pruning Methods

	Sparsity	TSA	Datasets of Fine-tuning	Structured		Unstructured
				Delete	Merge	
TSEP [19]	96.875%	S	GLUE [190] SQuAD [146]	✓		
NAEE [114]	50%	S	MetaMathQA [218]	✓		
UNCURL [157]	75%	S	FLAN [113]	✓		
PEMPE [22]	75%	A	CIFAR-10 CIFAR-100 [87] ImageNet [153]	✓		
SEER-MoE [123]	25%	S	MMLU [63] SST5 [170]	✓		
MoE-I^2 [210]	53.98%	A	Alpaca [178]	✓		
DEK [229]	75%	S	C4 [143]		✓	
EEP [106]	75%	A	NA	✓	✓	
MC-SMoE [101]	75%	S	eight datasets		✓	
MoE-Pruner [202]	50%	A	C4 [143]			✓
STUN [89]	70%	A	C4 [143]			✓
MoE-compression [60]	50%	A	Lima [235] MetaMathQA [218]	✓		✓

Sparsity indicates the max ratio of pruning. TSA represents task specific or agnostic. EEP [106] does not require additional fine-tuning. Finetuning datasets of MS-SMoE [101] are SST2 [170], MRPC [36], MultiRC [80], COPA [151], WinoGrande [154], SQuAD [146], WikiQA [212], and HotpotQA [213].

considering task-specific inference optimization can give guidance for later stages. PEMPE [22] prunes experts that have smaller changes in the router's l_2 norm between the pre-trained and fine-tuned models, and will conduct experiments with other compression techniques on SOTA vision MoEs. SEER-MoE [123] employs a heavy-hitters counting method for expert pruning, then with regularization-based fine-tuning reaches further expert pruning. MoE-I^2 [210] proposes the

Layer-wise Genetic Search and Block-wise KT-Reception Field with the non-uniform pruning ratio to prune unimportant experts.

In addition to direct pruning, some studies utilize expert merging to reduce the number of experts. For unstructured pruning, MoE-Pruner [202] prunes weights with the smallest magnitudes, weighted by the corresponding input activations and router weights. The pruned MoE models can benefit from a pre-trained teacher model through expert-wise knowledge distillation and have compatibility with structured pruning. Moreover, STUN [89] combines structured and unstructured pruning to achieve better performance than unstructured pruning alone, utilizing the sparse character of MoEs based on behavior similarity in a greedy manner. MoE-Compression [60] proposes a unified framework for MoE compression, using both structured and unstructured pruning methods to achieve significant inference speedup with minimal accuracy loss.

Another potential approach is merging outdated experts based on their parameters to achieve better performance. Branch-Train-Merge [99] independently trains different parts of the model on distinct subsets of data, eliminating the need for large-scale multi-node synchronization typically required in traditional LLM training. Building on this, Branch-Train-Mix [173] trains multiple copies of a seed LLM to specialize in multiple domains in an asynchronous and parallel fashion, then merges the parameters of the MoE layer to create a unified model that can be further trained. The second finetuning stage makes the final LLM more performant. They also find that their approach is more computing efficient compared to the dense model or specialized MoE model on training. Park et al. [136] observed that introducing a shared layer in the MoE could lead to performance degradation. To address this, they trained merged experts that learned the same features in different ways, improving their ability to generalize and mitigating catastrophic forgetting during incremental learning of multi-domain tasks. MC-SMoE [101] divides experts into different groups based on routing policies and then merges each group into one expert. HC-SMoE [18] is a framework that uses hierarchical clustering to merge experts without requiring retraining and can be applied in a task-agnostic manner. During inference, MEO [61] systematically investigates the computational cost of MoE. In the drop-in replacement algorithm, they first merge the selected expert parameters and then perform efficient computation. FoE [192] fuses the outputs of expert models by leveraging their complementary knowledge of the data distribution, framing it as a supervised learning instance. DEK [229] identifies and groups similar experts in feature space, then merges experts within the same group in weight space to reduce the expert count. EEP [106] introduces a two-stage approach to both prune and merge experts, reducing the total number of experts (saving GPU memory) and the number of active experts (accelerating inference). Inspired by the concept of knowledge transfer in multi-task learning, HyperMoE [232] proposes a novel expert network that further increases sparsity while utilizing information from unselected experts as supplementary input. In practice, they capture the contextual information of experts to compensate for the performance loss of transferring knowledge to specific experts. LiteMoE [242] retains the most critical experts based on the application's characteristics, merges secondary experts, and obtains the final sparse model without retraining. This approach enables efficient deployment of lightweight submodels on resource-constrained mobile devices.

3.2.2 Expert Quantization. In addition to model pruning, quantization is an effective technique for reducing model size by converting high-precision weights into low-precision. Given the redundancy of experts in MoE models, current research primarily focuses on quantizing the weights of the experts in MoE. As illustrated in Figure 4(b), experts are quantized into appropriate low-precision versions using various methods. MC-MoE [66] leverages the access frequency and activation weight to assess the importance of each expert. Along with the associated quantization loss, these metrics are used to determine the optimal quantization configuration for experts via the Integer

Table 3. A Comparison of Quantization Methods

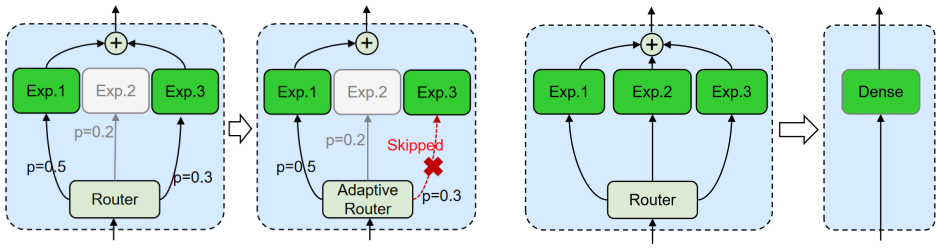
Method	Quantization Type		Memory Reduction	Accuracy Drop	Inference Speedup	Quantization Bits
	Weight	Activation				
MC-MoE [101]	✓		4.27x	3.8%	1.80x	1, 2, and 3
MoE-CSP [84]	✓		4.00x	-	26.00x	4, 8
MoQE [83]	✓		4.90x	0.97%	-	2, 3, and 4
QMoE [47]	✓		20x	6.7%	0.95x	1, 2
CMoE [220]	✓	✓	150x	23.81%	-	1, 2, and 4
MoE-MPTQS [72]	✓		-	0 ∼ 4.98%	1.00x ∼ 20.63x	4, 8
HOBBIT [177]	✓		-	1%	1.35x	2, 4
EdgeMoE [215]	✓		1.05x ∼ 1.18x	5%	1.11x ∼ 2.78x	2, 4, and 8

Programming model. Specifically, the ith expert's access frequency is defined as $\phi_i = \frac{n_i}{N}$, and the activation weight is defined as $w_i = \frac{\sum_{j=1}^{N} \sigma_j}{N}$, where n_i is the usage frequency, σ_j is the routing weight, and N represents the size of the calibration dataset. The quantization loss, ϵ_{ij} (computed using the Frobenius norm), is then determined for quantizing expert i to j bits ($j \in 1, 2, 3$). Using these metrics, MC-MoE defines the objective function as $\sum_{i=1}^{n} \sum_{j=1}^{3} \phi_i^{\alpha} \cdot w_i^{\beta} \cdot (\epsilon_{ij} \cdot x_{ij})^{\gamma}$, which is minimized using Integer Programming to determine the optimal bit-width for each expert. MoE-CSP [84] quantizes expert weights to either 4 or 8 bits to reduce memory consumption in MoE models. Additionally, it designs specific CUDA kernels that handle the 4-bit/8-bit quantized weights and perform floating-point calculations to accelerate computations. MoQE [83] quantizes expert weights to 2 bits to address the memory and latency challenges of MoE models based on its observations that quantizing expert FFN layers to 2 bits does not significantly affect model quality, while quantizing other components, like self-attention, significantly hurts performance. Further advancements include QMoE [47] and CMoE [220], which aggressively compress MoE model into just 1-bit. QMoE implements a highly scalable compression algorithm for large models and introduces a custom compression format along with bespoke GPU kernels for efficient on-the-fly computation. On the other hand, CMoE uses binary-weight networks to quantize model weights to 1-bit and applies learned step-size quantization to activations, quantizing them to 4 bits for MoE-based ASR models, enabling deployment on embedded devices. Moreover, some MoE-optimized systems leverage quantization for better system efficiency. MoE-MPTQS [72] and HOBBIT [177] dynamically select quantized experts to replace original experts based on the current inputs, thereby reducing the expert loading cost on memory-limited devices. EdgeMoE [215] statistically determines the appropriate expert bit-width by profiling expert importance on a calibration dataset. Furthermore, QMoE-Benchmark [100] provides a benchmark for exploring various MoE structure-aware quantization heuristics, from coarse to fine granularity. The study reveals that different MoE structures (e.g., blocks, experts, and linear layers) require different numbers of weight bits for effective and efficient quantization.

We summarize the main results reported by the methods discussed above in Table 3. From the table, we observe that quantization primarily benefits memory reduction, with most methods achieving more than a 4x reduction in memory usage. Some methods also lead to actual inference speedup, while others do not. For instance, QMoE even incurs a 5% overhead due to the absence of a dedicated 1-bit inference CUDA kernel implementation and hardware support. Furthermore, quantization typically causes some accuracy loss, with lower bit widths resulting in greater accuracy degradation. Therefore, when applying quantization to a model, it is crucial to strike a balance between memory consumption, accuracy, and inference speedup, taking into account the specific requirements and available hardware resources.

3.2.3 Expert Distillation. Knowledge distillation is another effective method for creating smaller, yet powerful models from larger ones. As shown in Figure 4(c), knowledge distillation presents a promising solution to generate a compact, high-performance model from the original MoE model. LLaVA-MoD [167] combines the MoE structure with knowledge distillation to efficiently train **small multimodal large language models** (**s-MLLMs**) from large ones (l-MLLMs). It first incorporates the MoE structure into the s-MLLM to balance computational efficiency and model performance. Then, it introduces two consecutive distillation stages, mimic distillation and preference distillation, to train the s-MLLM. Mimic distillation minimizes the **Kullback-Leibler** (**KL**) divergence between the output distributions of the s-MLLM and l-MLLM, enabling the s-MLLM to emulate the l-MLLM's understanding. Preference distillation further refines the s-MLLM using Preference Optimization with additional datasets. DeepSpeed-MoE [144] employs staged knowledge distillation to create a distilled version of its proposed PR-MoE model, called MoS. This method reduces model size while maintaining performance. Additionally, some studies focus on transferring the power of sparse models to dense models through knowledge distillation for more efficient deployment. For example, OneS [205] generates a dense model from a MoE model in two steps: knowledge gathering and knowledge distillation. Knowledge gathering merges experts into a single expert using four aggregation methods, including summation, averaging, top-k knowledge gathering, and **Singular Value Decomposition** (**SVD**) knowledge gathering. In the second step, knowledge distillation distills the merged model using the original MoE model. What's more, MoE-KD [155] and CMoE [220] distill MoE-based speech recognition models into dense models, accelerating the speech recognition process. Specifically, MoE-KD initializes the FFN module of the student dense model with the most frequently used expert from the teacher MoE model, then trains the FFN modules of the student model through layer-wise distillation. CMoE distills a binary dense model from the original model using quantization techniques. Switch Transformers [46] and ELSM [9] also explore distillation techniques to convert their sparse models into dense models. Additionally, to simplify the construction of MoE models, LaDiMo [81] converts a dense model into a sparse MoE model via layer-wise distillation, where each expert learns to approximate the results of the original dense layers.

3.2.4 Expert Decomposition. As shown in Figure 4(d), low-rank decomposition is an effective method for reducing model size by decomposing a large weight matrix into smaller matrices. MPOE [52] employs the **matrix product operator** (**MPO**), a tensor decomposition technique derived from quantum many-body physics, to decompose the expert weight matrix into a central tensor and several auxiliary tensors. The central tensor retains most of the parameters and core information of the original weight matrix, while the auxiliary tensors are much smaller and serve as complements to the central tensor. After decomposition, all experts in a layer share the same central tensor, thereby significantly reducing the total number of parameters in that layer. MC-SMoE [101] first groups the experts into several clusters and then merges each group into a single expert using a weighted sum. Low-rank decomposition is then applied to the merged experts to further reduce the model size. This approach is based on the observation that the merged experts have a lower rank compared to the original experts, making them more suitable for decomposition. MoE-I^2 [210] identifies the importance of each expert and assigns higher ranks to more important experts while assigning lower ranks to less important ones for low-rank decomposition. The importance, $I_{i,j}$, of the jth expert in the ith layer, $e_{i,j}$, is determined by removing $e_{i,j}$ and calculating the performance loss compared to the original model. The SVD rank, $r_{i,j}$, for $e_{i,j}$ is then computed as $r_{i,j} = \left\lfloor \frac{(I_{i,j}+\epsilon)^\alpha}{\sum_{j=1}^{M_i}(I_{i,j}+\epsilon)^\alpha} \right\rfloor \cdot R_a \cdot M_i$, where ϵ and α are hyperparameters, R_a is the compression ratio, and M_i is the number of experts in layer i.



(a) Experts Skipping (b) Sparse to Dense

Fig. 5. Algorithm improvement on expert layers.

3.3 Algorithm Improvement

In this part, we conduct an in-depth exploration of two other different strategies aimed at improving the MoE model inference algorithm.

3.3.1 Dynamic Gating. Given the significant progress made in utilizing the sparsity of MoE models, specific strategies can be employed to further exploit this sparsity in the inference process. Due to the vertical parallel structure of MoE experts, dynamically reducing the number of experts activated for each token clearly presents an effective strategy. Figure 5(a) illustrates the calculation process of MoE layer when the skip mechanism is applied, in contrast to the original process.

Li et al. [95] proposed a self-adaptive gating mechanism that dynamically determines the number of experts required for each token, based on the expert probability distribution. This method enhances training efficiency while preserving the sparsity of the MoE model and further reduces training time through curriculum learning. DynMoE [55] eliminates the need for fixed top-k hyperparameters by enabling each token to dynamically select how many experts to activate based on its complexity. It jointly optimizes a gating network that predicts both the number and identity of experts per token, reducing computational overhead while maintaining model performance. XMoE [211] employs a strategy of using smaller, but more experts, with dynamic expert activation based on threshold values to balance computational efficiency and model performance. Adap-MoE [234] dynamically adjusts the number of activated experts per layer based on its sensitivity to expert sparsification—reducing experts in insensitive layers while preserving them in accuracy-critical ones. This layer-wise adaptive gating cuts average expert usage by 25% without accuracy loss, and is complemented by expert prefetching and cache management to lower inference latency on edge devices. The comprehensive comparison of performance and methods of these works is presented in Table 4. DA-MoE [4] introduces a token importance prediction method derived from attention mechanisms to guide expert allocation. By assigning experts based on token importance scores, their approach achieves strong performance on the GLUE benchmark while demonstrating effective scaling with increased expert count.

3.3.2 Sparse to Dense. In certain scenarios, dense models offer unique advantages due to their smaller number of parameters. Therefore, transforming MoE models into dense target models all at once can achieve maximum sparsity while maintaining model performance. Most approaches use knowledge distillation to achieve this sparse-to-dense conversion, as illustrated in Figure 5(b).

XFT [35] proposes a novel method to supervise fine-tune dense LLMs. It first generate a sparse-upcycled MoE model, and then transform it back into an efficient dense LLM of the same size and structure through a learnable merging mechanism. Switch Transformers [46] explores distillation techniques to convert a sparse model into a dense one, demonstrating that the dense model can retain over 30% of its performance even after compressing 97% of the parameters from

Table 4. A Comparison of Dynamic Gating Methods

Method	FLOPs Reduction	Speedup	Threshold Strategy	Load Balance	PR.
Fixed top-k gating	0%	1.0x	✗	✗	✓
Li et al. [95]	38.2%	1.32x	accumulative probability	soft on top-1	✓
DynMoE [55]	9%	1.37x	single expert probability	✓	✗
XMoE [211]	75%	-	accumulative probability	✓	✓
AdapMoE [234]	25% of experts	1.35x	performance perturbation	✗	✓

PR. indicates performance Retention. Specially, Li et al. [95] only uses soft load balance constraints on top-1 gating. No specific acceleration ratio was provided in XMoE [211].

the sparse MoE model. ELSM [9] demonstrates that dense student models distilled from sparse MoE teachers can match and even surpass the teacher's performance. OneS [205] employs four distinct methods to generate the dense model, including summation, averaging, top-k Knowledge Gathering, and SVD Knowledge Gathering. The model is then refined to reduce noise by gathering sparse knowledge. TSEP [19] progressively eliminates non-expert components based on specific downstream tasks, ultimately converting the sparse MoE model into a dense counterpart. EWA [67] replaces FFNs with specially designed MoEs during training before reverting to dense ViT for inference. AdaMoLE [112], which combines the **Low-Rank Adaptation** (**LoRA**) structure, a dedicated network is used to adjust the activation threshold for different task complexities. It has shown superior performance to baseline in many natural language processing tasks, especially in some commonsense reasoning tasks. **expert-choice routing** (**ECR**) [238] exploits the relative importance of different tokens, so each expert in MoE block can have a fixed bucket size to avoid an expert being under-specialized or over-specialized. SoftMoE [138] introduces a fully differentiable sparse MoE architecture that replaces hard token-to-expert routing with implicit soft assignment. This design outperforms both dense Transformers and hard-routing MoEs in visual recognition tasks.

4 System-level Optimizations

Due to the unique structure of MoE, many studies focus on accelerating inference at the system level by leveraging the sparse activation patterns inherent to this architecture. Typically, MoE models are deployed in two scenarios: cloud environments with multiple powerful servers and edge environments with a single device. In cloud clusters, MoE models are distributed across many devices to enable parallel execution. In addition to traditional parallelization techniques such as data parallelism, tensor parallelism, and pipeline parallelism [74, 124, 145], expert parallelism is a specialized method tailored specifically for MoE models. On edge devices, limited GPU memory often cannot accommodate all parameters of an MoE model, necessitating the offloading of some parameters to CPU memory or SSD storage. To address this, the expert-offloading technique has been developed to fully utilize the sparse activation patterns of experts for efficient execution. Figure 6 shows the detailed structure of this section.

4.1 Expert Parallelism

Expert parallelism is a key technique for deploying large MoE models across multiple devices to enable distributed execution. As Figure 7(a) shows, when distributing MoE layers, each device holds a subset of experts along with all other non-expert parameters [166]. In special cases where an expert is very large, each device may contain only one expert. During the execution of an MoE layer, inputs are processed on each device by the attention module and gate module. Subsequently, an all-to-all communication operation redistributes tokens to corresponding devices based on the gate output, allowing each expert to process its assigned inputs. Finally, another all-to-all communication operation occurs to exchange the experts' outputs and send them back to the

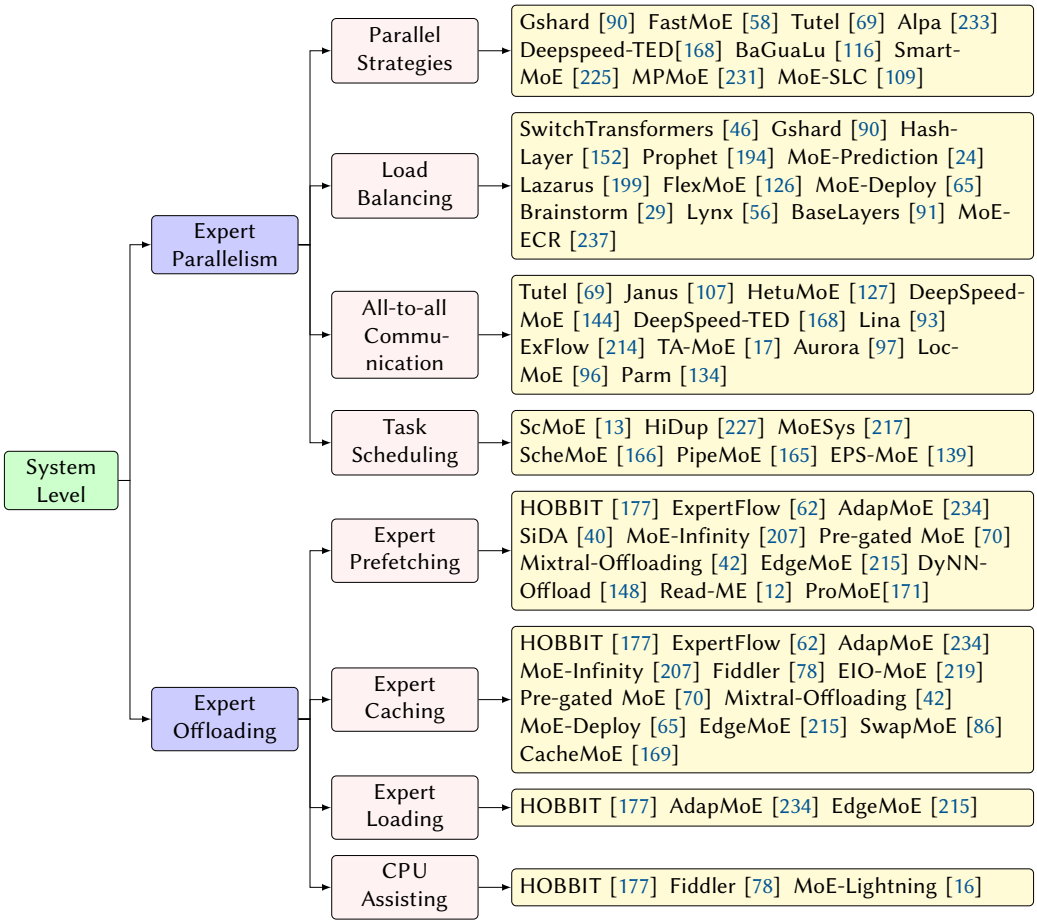
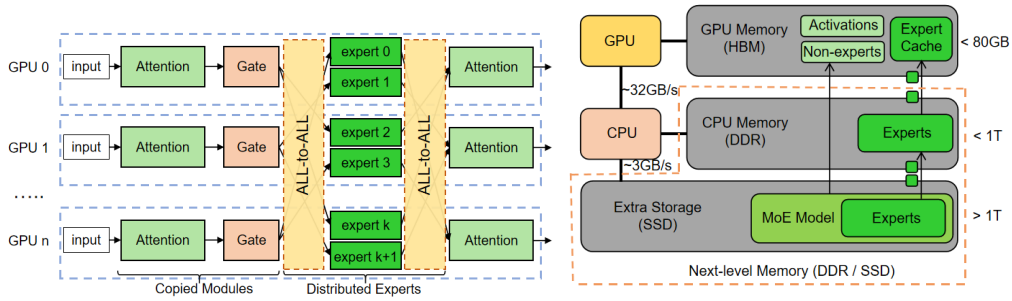


Fig. 6. System-level inference optimization techniques for MoE.

originating devices. Consequently, the execution time of MoE layers is dominated by two primary phases: computation and communication. To accelerate inference, various studies have proposed optimizations for these phases.

Existing research focuses on optimizing MoE training and inference performance along four main dimensions: parallelism strategy, load balancing, all-to-all communication, and task scheduling.

4.1.1 Parallelism Strategy Design. In addition to relying solely on expert parallelism [58, 90], recent studies combine multiple parallel strategies to maximize parallelism and achieve more efficient distributed computing. Tutel [69] dynamically switches parallelism strategies at each iteration without incurring extra overhead. Specifically, this work employs a single distribution layout that encompasses all optimal strategies, thereby eliminating the need to reformat input data or weights when switching parallelism strategies. Alpa [233] reclassifies conventional data and model parallelism into intra-operator and inter-operator parallelism based on the key observation that different parallelization techniques have varying bandwidth requirements for communication. In typical clusters, co-located devices have high communication bandwidth, whereas distant devices have limited communication bandwidth. With this reclassification, Alpa can automatically derive efficient parallel execution plans that incorporate data, operator, and pipeline parallelism based



(a) Expert Parallelism (b) Expert Offloading

Fig. 7. Expert parallelism and expert offloading techniques.

on the model description and cluster configuration. DeepSpeed-TED [168] implements a hybrid parallel algorithm combining data, tensor, and expert parallelism. It designs a tiled version of the optimizer to reduce peak memory consumption arising from the combination of these three parallelism forms, thereby enabling the training of MoE models with 4–8× larger base models than the current state-of-the-art. BaGuaLu [116] proposes a parallel strategy called MoDa that combines MoE parallelism and data parallelism to effectively scale models. Specifically, MoDa places data parallelism within a single supernode while maintaining MoE communication globally across supernodes. This approach is based on the observation that All-reduce communication (in data parallelism) incurs much higher costs than All-to-all communication (in MoE parallelism) during the training of its target large-scale models. SmartMoE [225] explores the space of hybrid parallelism with an awareness of heterogeneous workloads, incorporating potentially faster candidate parallel strategies. It introduces a two-stage approach to identify optimal parallel strategies for data-sensitive models, periodically determining whether a new parallel strategy should be employed at runtime. MPMoE [231] proposes a lightweight profile-based algorithm that leverages profiling information to identify the most suitable configuration at runtime, optimizing pipeline parallelism. Additionally, in serverless computing, MoE-SLC [109] proposes a Bayesian optimization framework with multi-dimensional δ-greedy search to learn expert selections for expert parallelism and optimize the MoE deployment to achieve optimal billed cost. By enforcing load balancing at the global-batch (corpus) level rather than per-sequence, global-batch LBL [140] enables experts to specialize by domain during training. At inference time, this leads to more consistent and accurate routing decisions, improving both model quality and computational efficiency without increasing FLOPs.

4.1.2 Load Balancing. Token processing in MoE models can lead to imbalances, where some experts receive significantly more tokens than others. This disparity results in certain devices experiencing high overheads while others remain idle, ultimately delaying overall computation. To address this issue, some works propose adding auxiliary load balancing loss terms to train the gate module [46, 90, 160], while others implement gate modules with special hash functions to prevent load imbalance [152]. Advanced techniques focus on optimizing expert placement. For example, Prophet [194] builds a load balance performance model to estimate the execution time of the MoE layer for each expert placement and then uses a greedy search algorithm to find a well-balanced expert placement. MoE-Prediction [24] traces and analyzes the loads of each expert during the training process, then employs classical prediction algorithms to forecast the expert load proportions, providing valuable guidance for expert placement in MoE training. Lazarus [199] allocates expert replicas using an optimal placement algorithm that assigns more replicas to popular

experts to help balance the load. FlexMoE [126] employs fine-grained replication strategies that select specific heavy experts and duplicate them across multiple devices, dynamically adjusting the expert-to-device mapping at runtime to address routing imbalances. Additional strategies, such as those proposed by MoE-Deploy [65] and Brainstorm [29], rely on historical allocation data and expert reordering to balance loads. Furthermore, techniques like Lynx [56] reduce the number of active experts during batch inference, and BaseLayers [91] formulates token-to-expert allocation as a linear assignment problem to ensure equitable token distribution. Additionally, MoE-ECR [237] allows experts to select the top-k tokens instead of letting each token choose the top-k experts, ensuring that each expert has a fixed bucket size.

4.1.3 All-to-all Communication Optimization. All-to-all communication is a significant bottleneck in MoE layer execution [59, 97, 144], prompting substantial research aimed at optimizing this operation. Many efforts focus on hierarchical communication strategies and data compression techniques to reduce overhead. For instance, Tutel [69], HetuMoE [127], and DeepSpeed-MoE [144] utilize the hierarchical all-to-all algorithm to optimize communication by combining intra-node and inter-node hierarchies. Compared to traditional all-to-all operations, which allow all GPUs to communicate directly with one another, this hierarchical approach first gathers data from all GPUs within a node to a single GPU (intra-node) and then communicates between nodes (inter-node). This method speeds up communication by fully utilizing both levels of bandwidth. Data compression strategies, such as those employed by DeepSpeed-TED [168] and TA-MoE [17], minimize data transfer by eliminating unnecessary information and adapting data volume to the network topology. Innovative paradigms like Janus [107] adopt a data-centric approach, moving experts between devices instead of tokens, thereby significantly reducing communication costs. ExFlow [214] reduces the number of all-to-all operations from two to one by ensuring token context coherence, meaning that all GPUs have the context of all requests rather than only their own. It then exploits inter-layer expert affinity to further reduce the number of all-to-all operations by optimally placing experts in each layer, increasing the probability that tokens remain on local GPUs. Aurora [97] orders token transmission to avoid bandwidth contention, achieving minimal all-to-all communication costs through theoretical derivation. LocMoE [96] and Parm [134] further enhance communication efficiency by converting inter-node communication to intra-node operations and overlapping intra-node with inter-node communications. Additionally, Lina [93] accelerates all-to-all communication by prioritizing all-to-all operations over all-reduce operations.

4.1.4 Task Scheduling. To overlap communication and computation tasks and reduce end-to-end runtime, various task scheduling strategies have been developed. Advanced scheduling strategies leverage architectural and algorithmic innovations to achieve this overlap effectively. For example, ScMoE [13] introduces a shortcut-connected MoE architecture that processes representations from both the current and preceding layers, rather than solely processing representations from the current layer. Specifically, ScMoE employs a top-1 MoE module to handle representations from the preceding layer via a shortcut connection, while a shared expert processes the current layer's representations. This structure effectively decouples communication from its conventional sequence, thereby enhancing the overlap between communication and computation. HiDup [227] splits the input data on each device into two microbatches with no dependencies between them. Consequently, the training of the two microbatches on each device can be carried out in parallel, effectively overlapping the computation of one microbatch with the communication of the other. MoESys [217] employs an elastic MoE training strategy with 2D prefetching and fusion communication over hierarchical storage to overlap computation with the parameter readiness from the hierarchical storage. ScheMoE [166] first modularizes the computing tasks (data compression and expert computation) and communication tasks (collective communication). It then designs

Table 5. Speedup of Parallelism Systems When Compared to DeepSpeed-MoE and FasterMoE in MoE Training

DeepSpeed-MoE [144]			FasterMoE [59]		
Method	Speedup	Metric	Method	Speedup	Metric
HetuMoE [127]	5.88x	Time per iteration	TA-MoE [17]	1.39x	Tokens per second
TA-MoE [17]	1.31x	Tokens per second	PipeMoE [165]	1.69x	Time per iteration
FlexMoE [126]	1.70x	Time to converge	FlexMoE [126]	1.30x	Time to converge
Prophet [194]	2.39x	Time per iteration	Prophet [194]	1.43x	Time per iteration
Parm [134]	3.00x	Time per iteration	ScheMoE [166]	1.25x	Time per iteration
Lazarus [199]	3.40x	Tokens per second	MPMoE [231]	1.60x	Time per iteration
DeepSpeed-TED [168]	1.35x	Time per iteration	SMARTMOE [225]	1.88x	Time per iteration

Table 6. Speedup of Parallelism Systems in MoE Inference

Method	Speedup	Metric	Baseline	Optimized Techniques
Lina [93]	1.63x	Time per batch	DeepSpeed [120]	Prioritize all2all over allreduce
TUTEL [69]	2.03x	Time per batch	Fairseq [149]	Design an identical layout for weights
Aurora [97]	1.81x	Time per batch	Lina [93]	Strategically order token transmissions
ExFlow [214]	2.2x	Tokens per second	Deepspeed-MoE [144]	Reduce two all2all operations into one
Brainstorm [29]	3.33x	Time per batch	Tutel [69]	Preload weight with profiling statics
MoE-Deploy [65]	3.32x	Tokens per second	Tutel [69]	Combine high-used experts with lows
DeepSpeed-MoE [144]	7.25x	Time per batch	PyTorch [5]	Grouped tokens with same data path

an adaptive optimal scheduling algorithm to pipeline the communication and computing tasks based on these modularized operations. Similarly, PipeMoE [165] designs a performance model to predict the computation and communication costs for various MoE workloads and then proposes an optimal polynomial-time solution to pipeline tasks, thereby hiding communication latency based on the performance model's results. Additionally, EPS-MoE [139] dynamically selects optimal kernel implementations to adaptively overlap FFN module computation with all-to-all communication, further improving runtime efficiency.

4.1.5 Performance Summarization. To assess the performance of the methods discussed above, we extract relevant data from their respective papers. Table 5 shows the training speedup of various systems relative to DeepSpeed-MoE and FasterMoE, both of which are widely recognized by researchers and serve as excellent starting points for related studies. Although different systems evaluate their methods using various configurations, a rough comparison can still be made due to the shared baselines. For instance, HetuMoE demonstrates exceptional performance, with the highest speedup compared to other systems. In addition to training speedup, we also report inference speedup and highlight the main optimization techniques of several systems in Table 6. However, comparing performance across systems is challenging due to differences in their baselines. Beyond speedup, some systems offer additional benefits, such as reduced memory usage. As shown in Table 7, many systems focus on optimizing memory usage, which is critical given the large size of MoE models. Additionally, some studies also optimize GPU utilization and communication efficiency.

4.2 Expert Offloading

When deploying MoE models on edge devices, parameter-offloading techniques offer a potential solution to mitigate the challenge of insufficient GPU memory for storing all model parameters. However, traditional parameter-offloading methods load model parameters layer by layer from CPU memory or SSD, neglecting the sparse activation characteristics of MoE models. This oversight incurs significant overhead in parameter loading, leading to suboptimal inference performance.

Table 7. Additional Benefits of Existing Systems

Method	Benefits	Baseline
Lina [93]	Gain 2.3x reduction in all2all communication	DeepSpeed [120]
Prophet [194]	Improve the load balance by 1.75x to 12.06x	FasterMoE [59]
ScheMoE [166]	Gain 1.4x to 2x speedup in all2all communication	NCCLA2A [127], 2DH-A2A [69]
MPMoE [231]	Achieve a 53% decrease in memory usage	FasterMoE [59]
MoESys [217]	Achieve a 18% decrease in memory usage	DeepSpeed [120]
Aurora [97]	Achieve a 1.57x to 1.72x increase in GPU utilization	Lina [93]
MoE-Deploy [65]	Achieve a 79.6% decrease in memory usage	Tutel [69]
DeepSpeed-TED [168]	Support 1.09x to 4.8x larger MoE models	DeepSpeed-MoE [144]

To address this limitation, many studies have proposed expert-offloading techniques, which take advantage of MoE sparse activation patterns. Instead of loading all experts for a given layer, expert-offloading selectively loads only the required experts, thereby significantly reducing loading latency.

As Figure 7(b) shown [177], expert-offloading operates by storing all non-expert weights and a subset of critical experts in GPU memory (referred to as the "expert cache") while offloading less frequently used experts to CPU memory or SSD (referred to as "next-level memory"). When required experts are missing from the expert cache, they are fetched from next-level memory and loaded into the cache, potentially replacing some existing experts. Various studies have optimized this process using different strategies to achieve superior inference performance.

4.2.1 Expert Prefetching. To minimize the waiting time for the required experts, some studies focus on optimizing prefetching techniques that overlap expert loading with GPU computation. Many works predict the experts needed for subsequent layers. For example, Mixtral-Offloading [42], AdapMoE [234], and HOBBIT [177] use current gating inputs to feed the gating module for the following layers, predicting the required experts and prefetching them in advance. This prediction is based on the key observation that the inputs to the gating function in the MoE module share high similarity across successive layers, due to the residual structure of LLMs. As reported in these works, the prediction accuracy can reach about 90%, significantly enhancing the effectiveness of this approach. Pre-gated MoE [70] introduces a pre-gated MoE structure, which identifies the experts required by the next layer during the computation of the current layer and preloads them accordingly. Although this new structure is suitable for prefetching, it may cause a decrease in model accuracy. EdgeMoE [215] constructs a prediction table using a calibrated dataset, leveraging the experts from the current layer to predict the experts for the next layer. This table is built on the observation that the expert activations of sequential layers are statistically correlated. This means that given prior knowledge of the expert activations from layers 1 to n-1, we can estimate the activation probability of each expert at the *n*th layer with high confidence. DyNN-Offload [148] trains a pilot model for each layer to predict the required experts for the next layer. MoE-Infinity [207] tracks the request-level usage frequency of each expert and prefetches high-priority experts based on this frequency. It prefetches experts across multiple layers concurrently, rather than just the next layer, but prioritizes the experts closest to the currently executed layer. ProMoE [171] uses a learned predictor to prefetch experts in a sliding-window manner. This predictor, a small neural network (a two-layer MLP), learns the correlation between layer inputs and expert selections. It can use the inputs of the *i*th layer to predict the expert selections of the (i+k)-th layer, where k is the sliding window size (e.g., 4 or 8). ProMoE runs LLM inference for multiple iterations to collect training data for the predictor in the offline stage and runs the predictor on the CPU to hide its execution latency by overlapping it with the LLM inference during the online stage. Additionally, some works aim to predict all the required experts for the current forward pass at the

start of the forward iteration. ExpertFlow [62] improves MoE inference efficiency by predicting the full routing path before computation begins, enabling accurate expert prefetching and adaptive CPU-GPU caching that minimizes I/O overhead. It further reduces expert activation overhead via dynamic token scheduling, which reorganizes input tokens across batches to maximize expert reuse and lower memory pressure. SiDA [40] employs a network-based hash function to predict the activated experts for each token at each layer for the current incoming batch of data. This predictor is constructed with an LSTM and trained using truncated knowledge distillation. Similarly, Read-ME [12] constructs a pre-gating router decoupled from the MoE backbone to predict all the required experts at once.

4.2.2 Expert Caching. Since GPU memory can only hold a subset of experts (the expert cache), optimizing the cache management policy is crucial for improving the cache hit ratio and reducing transfer time. Several works [42, 65, 70, 215, 219] adopt the **Least Recently Used** (**LRU**) policy, based on the assumption that recently used experts are more likely to be accessed again. According to Mixtral-8x7B [75], when an expert is selected for the ith token, it has a significantly higher probability (more than 10% in most layers) of being selected again for the (i+1)-th token compared to a random selection. In contrast, MoE-Infinity [207] employs a variant of the **Least Frequently Used** (**LFU**) policy to define the caching priority of experts, leveraging its request-level tracking capability. Fiddler [78] maintains a set of important experts in GPU memory using a static dataset to profile the active times of experts, treating these times as a measure of their importance. This method demonstrates an improvement of approximately 3 percentage points in hit rate compared to random placement. Similarly, SwapMoE [86] stores a set of important experts in GPU memory based on a defined expert importance score, but it dynamically updates the cached experts at a suitable frequency. AdapMoE [234] introduces a dynamic cache size for different model layers, which is driven by its adaptive gating algorithm. This algorithm selects a varying number of experts for different layers, thus adjusting the cache size accordingly. Specifically, it assigns a larger cache size to layers that select more experts or experience lower prediction accuracy. HOBBIT [177] proposes a multi-dimensional cache policy that combines LRU, LFU, and a novel **Least High Precision Used** (**LHU**) strategy for its mixed-precision expert cache. Since high-precision experts incur higher loading latencies than low-precision experts, the LHU policy prioritizes high-precision experts to reduce loading costs. Additionally, CacheMoE [169] employs a cache-aware strategy that adjusts the router logits $z = G(x)$ to enhance the likelihood of selecting experts already present in the cache. This approach effectively increases the expert cache hit ratio, thereby reducing the costs associated with expert loading.

4.2.3 Expert Loading. Expert loading time often constitutes the primary bottleneck in expert-offloading inference. As reported in HOBBIT [177], expert loading consumes more than 80% of the total inference time. To address this, several works aim to directly reduce loading costs. For instance, EdgeMoE [215] and HOBBIT [177] reduce loading times by using low-precision experts instead of high-precision ones. Low-precision expert loading requires significantly less time than high-precision experts and can still maintain model accuracy, provided that only the less important experts are replaced. EdgeMoE determines the precision of each expert by profiling its importance using a static dataset. However, this method is highly dependent on the profiled dataset, which reduces its flexibility across different environments and may negatively affect accuracy. To address this limitation, HOBBIT introduces a token-level dynamic expert loading mechanism. This mechanism dynamically selects the appropriate precision for cache-missing experts based on the current inputs, ensuring both accuracy and flexibility. Specifically, it computes the importance score of a cache-missing expert based on the outputs of the gating module. If the score is below a predefined threshold, the system fetches a low-precision version; otherwise, it retrieves the

Table 8. Speedup of Offloading Systems Evaluated on Mixtral-8x7B model [75]

Method	Speedup	Metric	Baseline	Main Techniques
ProMoE [171]	1.16x	Tokens per second	LRU-MoE [171]	Train a predictor to prefetch experts
Fiddler [78]	8.20x	Tokens per second	Mixtral-Offloading [42]	Use CPU to help computation
HOBBIT [177]	2.30x	Tokens per second	MoE-Infinity [207]	Load adaptive precision experts
AdapMoE [234]	1.36x	Latency per sample	Mixtral-Offloading [42]	Skip unimportant experts dynamically
ExpertFlow [62]	1.99x	Tokens per second	Cache-MoE [62]	Train a predictor to prefetch experts
MoE-Infinity [207]	1.96x	Latency per token	Mixtral-Offloading [42]	Fetch experts with request-level tracing
MoE-Lightning [16]	3.50x	Tokens per second	FlexGen [164]	Schedule CPU-GPU-I/O pipeline
Mixtral-Offloading [42]	2.28x	Tokens per second	Accelerate [68]	Use LRU to cache important experts

Table 9. Speedup of Offloading Systems Evaluated on Switch Transformer model [46]

Method	Speedup	Metric	Baseline	Main Techniques
EdgeMoE [215]	2.82x	Latency per token	IO-EXP [215]	Quantize experts into different precisions
SwapMoE [86]	2.00x	Latency per sample	Original-MoE [86]	Update cached experts dynamically
SIDA [40]	3.93x	Samples per second	DeepSpeed [120]	Prefetch experts with a hash function
ExpertFlow [62]	5.23x	Tokens per second	SE-MoE [161]	Train a predictor to prefetch experts
MoE-Infinity [207]	4.53x	Latency per token	Accelerate [68]	Prefetch experts with request-level tracing
Pre-gated MoE [70]	1.50x	Tokens per second	MoE-OnDemand [70]	Design a pre-gate MoE structure

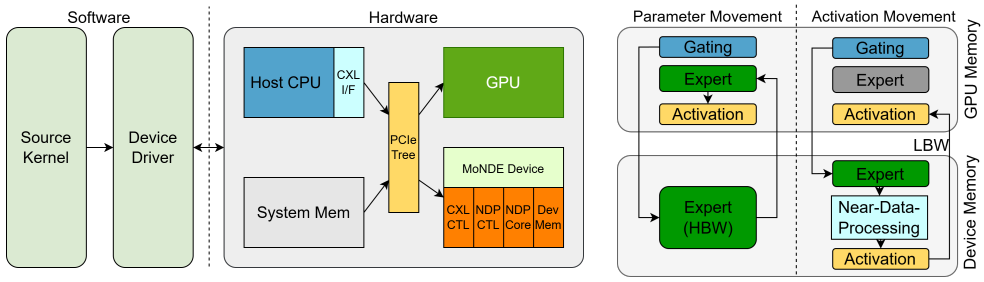
high-precision version to preserve model accuracy. Additionally, AdapMoE [234] employs an adaptive gating algorithm that skips less important experts. This algorithm uses the Fisher information matrix to compute the importance of each expert, further lowering expert loading costs.

4.2.4 CPU Assisting. Beyond relying solely on the GPU for computation, some approaches leverage the CPU to assist in the process. Fiddler [78] performs MoE computation on the CPU by copying intermediate activation values from GPU memory to CPU memory when the required experts are unavailable in GPU memory and then transferring the outputs back to the GPU. This method eliminates the need to load missing experts into GPU memory. As measured, the latency for loading an expert from CPU memory to GPU memory is significantly higher than the combined latency of copying activation values to the CPU, performing expert computation on the CPU, and then copying the results back to the GPU. HOBBIT [177] follows a similar pattern, using the CPU for computation with low-precision experts. Additionally, MoE-Lightning [16] introduces an innovative CPU-GPU-I/O pipelining strategy that enables the simultaneous utilization of both CPU and GPU resources, thus enhancing overall system efficiency. However, this approach is only applicable when CPU resources are sufficiently available. It may not be suitable for platforms with limited CPU resources or shared memory devices, such as the Jetson Orin.

4.2.5 Performance Summarization. Mixtral-8x7B [75] and the Switch Transformer [46] are popular MoE models, and most offloading systems use them as evaluation workloads. We extract relevant performance data from various papers to assess their effectiveness. Table 8 shows that many studies use Mixtral-Offloading as the baseline when evaluating the Mixtral-8x7B model. From the table, it is evident that Fiddler achieves a significant speedup compared to other systems, as it effectively utilizes the CPU to assist computation in GPU-limited environments. However, Table 9 shows that the methods when evaluated on Switch Transformer use different baselines, making direct comparisons difficult. Therefore, for our own studies on this topic, Mixtral-8x7B should stand out as a more suitable workload to consider.

4.3 Framework

Most of the systems mentioned above are built on top of popular deep learning frameworks such as PyTorch [5], Transformers [68], and DeepSpeed [120]. The data is summarized in Table 10. From the table, it is evident that most works in the expert parallelism category implement their systems based



(a) Overview of MoNDE

(b) Parameter and Activation Movement

Fig. 8. (a) MoNDE [82] system design integrating software and hardware. (b) The data flow between AM and PM.

Table 10. Statistics of the Number of Systems Built on Popular Current Frameworks

	Pytorch [5]	Transformers [68]	DeepSpeed [120]	Fairseq [149]	Llama.cpp [53]	vLLM [88]	FasterTransformer [129]
Parallelism System	12	5	9	2	0	0	0
Offloading System	4	7	1	0	2	1	1

on PyTorch, DeepSpeed, and Transformers. PyTorch serves as the foundational deep learning framework, with other libraries like Transformers and DeepSpeed built on top of it. As a result, directly building a system based on PyTorch offers greater flexibility in designing custom functions, though it lacks some popular techniques found in other libraries, such as model parallelism. DeepSpeed provides numerous features to assist in model training, particularly in distributed environments. Transformers, on the other hand, offers many pre-trained models, allowing developers to easily modify the model structure within the library. While this is very convenient for quickly testing ideas, it may not provide the best performance in terms of throughput and latency.

In the expert offloading category, most works still rely on Transformers due to its ease of use. However, some works also utilize Llama.cpp [53], vLLM [88], and FasterTransformers [129]. These three frameworks are optimized for deploying LLMs and offer great performance, but they tend to be more complex to use.

In conclusion, if you are looking to quickly verify an idea, Transformers is an excellent choice. For training models in distributed environments, DeepSpeed is the better option. If your focus is on designing an inference system with high performance, Llama.cpp and vLLM are good starting points. Otherwise, PyTorch remains a solid choice.

5 Hardware-Level Optimization

Recent hardware optimizations for MoE inference have addressed key challenges through novel architectures and co-design approaches. These optimizations target critical issues such as **operations per byte** (**Op/B**) efficiency, heterogeneous computing units, and memory access patterns. The following discusses significant advances in hardware-level solutions.

MoNDE [82] introduces a **near-data processing** (**NDP**) solution to address the challenges of sparse activation and expert parameter transmission overhead (Figure 8). The architecture integrates **Compute Express Link** (**CXL**)-based NDP controllers with specialized NDP cores for in-memory computation, utilizing LPDDR SDRAM (low power double data rate synchronous dynamic random-access memory) for high bandwidth and energy efficiency. The system implements a hybrid computing strategy where GPUs handle frequently accessed "hot" experts while NDP units

process "cold" experts, enabling concurrent execution through an Activation Movement paradigm rather than traditional Parameter Movement pattern.

FLAME [105] is the first accelerating framework that fully exploits MoE sparsity for transformers on FPGA. At the parameter level of the model, it utilizes M:N pruning to reduce unnecessary calculations, which can make a balance between column-balanced structured pruning and unstructured pruning. At the expert level, sparse activation prediction is performed through **circular expert prediction** (**CEPR**). By changing the patterning of the activation path of experts, the accuracy of expert predictions can be effectively improved. Then, using a double buffering mechanism to load the predicted expert while computing the previous expert to improve expert deployment efficiency.

M^3ViT [44] and Edge-MoE [156] constructed their FPGA architecture based on the reordering of attention computation in multitasking scenarios. For inference, M^3ViT can activate only the sparse "expert" pathway relevant to the task of interest for efficiency, and can further achieve zero-overhead switching between tasks with their hardware-level co-design. Edge-MoE is the first end-to-end FPGA implementation for multi-task ViT proposed some aggressive techniques, including an approximate method for solving the excessive complexity of GELU function calculation on FPGA and a unified linear layer module to achieve efficient reuse of hardware resource.

Duplex [223] selects a destination suitable for each layer execution in the device that combines xPU and Logic **processing-in-memory** (**PIM**). That means it could integrate two types of processing units that share device memories. Due to the bottleneck in computation and memory access between these two processing units which can complement each other, high computation and memory access utilization can be achieved simultaneously on the same device. Besides, it introduced an alternative PIM microarchitecture. The logic PIM optimized low Op/B operations through powerful processing units on the logic die, as well as more **through silicon vias** (**TSVs**) to achieve high bandwidth communication between the DRAM die and the logic die. Furthermore, it could execute expert and attention stages in parallel to maximize inference efficiency.

Space-mate [135] has provided its accelerator design for **simultaneous localization and mapping** (**SLAM**) tasks on mobile devices. Mainly including an **Out-of-Order** (**OoO**) SMoE router to alleviate data transactions for low latency, a **single skip** (**SS**) and **dual-skip** (**DS**) heterogeneous core architecture to exploit coarse-grained sparsity caused by similar zero patterns in the same expert for high throughput and energy-efficiency.

6 Future Directions and Open Challenges

Despite significant advances in MoE inference optimization, critical challenges and opportunities exist throughout the computing stack. This section analyzes future research directions along three interconnected dimensions: (1) computing infrastructure, from hardware to system software; (2) key system requirements, including performance, reliability, and efficiency; and (3) the development support ecosystem.

6.1 Computing Infrastructure Optimization

6.1.1 Hardware Integration and Acceleration. Efficient MoE execution demands hardware that transcends traditional dense-computation paradigms [44, 135, 156, 223]. Current platforms struggle with the sparse and dynamic nature of MoE workloads, necessitating specialized hardware solutions [174, 216, 236]. Immediate opportunities include developing specialized circuits for expert routing, creating memory hierarchies optimized for sparse parameter access, and implementing hardware support for dynamic workloads to reduce overhead and improve performance.

Beyond traditional optimizations, emerging platforms offer exciting possibilities. Neuromorphic systems, with their inherent support for sparse, event-driven computation, are a natural fit for accelerating expert activation patterns [158, 159]. Similarly, quantum computing might offer novel

approaches to complex routing decisions, and new memory technologies could provide more efficient storage and access for expert parameters, though significant research challenges remain in bridging classical and quantum computations [103, 132, 172].

6.1.2 System Software Optimization. The system software stack presents significant optimization challenges, as current operating systems and runtimes lack native support for the sparse computation and dynamic scheduling required by MoE models [29]. This gap necessitates fundamental innovations in system software.

Memory management systems require substantial adaptation to accommodate the dynamic expert activation in MoE models. Traditional virtual memory and cache hierarchies, which are optimized for the regular access patterns of dense LLMs, are poorly aligned with the highly irregular and input-dependent memory behavior of MoE architectures [88]. Future work should therefore investigate specialized memory management techniques capable of effectively predicting and prefetching expert parameters, potentially through new abstractions for sparse tensor operations. In distributed environments, resource allocation is another critical area. The dynamic nature of expert activation complicates placement and migration, requiring robust mechanisms for load balancing, fault tolerance, and resource elasticity [24, 194].

Ultimately, the co-design of hardware and software is crucial. Success requires close collaboration between hardware architects and algorithm designers to ensure that optimizations at both layers align with MoE computational requirements, minimizing energy consumption, and data movement while maximizing performance.

6.2 Operational Challenges and Requirements

Deploying and operating MoE systems in production environments introduces complex challenges spanning performance optimization, system reliability, and resource efficiency. These often-conflicting requirements must be carefully balanced for practical adoption across different application domains.

6.2.1 Energy Efficiency and Sustainability. Energy efficiency has received less attention than throughput and latency in MoE research, but the growing environmental impact of AI necessitates making it a primary optimization objective [7, 26, 64, 117, 147]. MoE models present unique energy challenges; while sparse activation reduces computation, overhead from routing, communication, and load balancing can lead to significant energy costs, especially on hardware optimized for dense workloads [69, 126, 128, 166, 240].

Future research should pursue carbon-aware deployment strategies, where expert placement and scheduling decisions consider the carbon intensity of data centers [43, 50, 142]. To support this, comprehensive energy accounting frameworks are essential. Current methods often ignore indirect costs like data movement and cooling; future energy models must capture the full environmental impact to enable truly informed optimization.

6.2.2 Latency and Quality of Service. The dynamic nature of MoE inference makes it difficult to provide consistent latency guarantees and maintain high availability, unlike traditional models with fixed computation paths [29, 207]. A fundamental challenge is the unpredictable, input-dependent expert selection, which causes variability in processing time. Future research must develop techniques to manage this variability, such as advanced caching, workload characterization, and adaptive routing algorithms that balance accuracy with computational overhead [62, 171, 234].

In distributed deployments, system reliability and fault tolerance are also critical [15]. The failure of an expert or communication link can degrade performance and quality. Opportunities exist for developing robust failure detection and recovery mechanisms, graceful degradation strategies, and

redundancy schemes that balance reliability with resource efficiency to maintain service quality under adverse conditions.

6.3 Development Support Ecosystem

6.3.1 Open-Source Frameworks. Current deep learning frameworks like PyTorch [5] and TensorFlow [180] lack native optimizations for MoE workloads, creating a significant barrier to efficient development. To address this, core frameworks require fundamental enhancements, including specialized operators for expert routing, efficient sparse computation primitives, and optimized memory management for expert parameters [111, 139]. These features must be deeply integrated to achieve performance comparable to dense model optimization.

To make MoE development more accessible, high-level APIs and abstractions are also essential. Current implementations often require deep expertise, limiting wider adoption [53]. Future frameworks should provide intuitive APIs for defining experts and routing, alongside seamless integration with the broader ML ecosystem. This involves creating standardized interfaces and compatibility layers so MoE systems can leverage existing tools for training, debugging, and deployment.

6.3.2 Benchmarking and Standardization. The rapid advancement of MoE optimization has created a pressing need for comprehensive benchmarking and standardization [49, 125]. Existing LLM benchmarks are ill-suited for MoE systems, as they fail to account for their unique architectural characteristics. This has led to fragmented evaluation practices that make fair comparisons difficult and impede systematic progress [49].

An effective benchmarking framework must be multi-dimensional, evaluating performance metrics like latency and throughput alongside model quality across various tasks and deployment scenarios. However, developing such standards is challenging due to the dynamic nature of MoE models and the need to balance trade-offs between objectives like quality and efficiency. Looking forward, the field requires standardized benchmark suites, consistent evaluation methodologies, and reference implementations to facilitate comparative analysis and accelerate research through more rigorous and comparable evaluations.

7 Conclusion

This survey provides a comprehensive overview of inference optimization techniques for MoE models across different abstraction levels. We have systematically analyzed various approaches, from model-level optimizations including efficient expert design, compression techniques, and algorithm improvements, to system-level solutions for distributed computing and expert offloading, and finally to hardware-level acceleration designs. Through this analysis, we observe that while MoE models offer a promising approach to scale model capacity with controlled computational costs, their efficient deployment requires careful consideration and optimization at multiple levels of the system stack. The field has seen rapid advancement with numerous innovative solutions, yet several challenges remain, particularly in areas such as hardware integration, energy efficiency, and standardized benchmarking.

Looking forward, we anticipate continued evolution in MoE inference optimization, driven by both academic research and industrial applications. The increasing adoption of MoE architectures in large-scale language models and multimodal systems will likely spur further innovations in optimization techniques. Key areas for future research include the development of specialized hardware architectures for sparse computation, more efficient expert routing algorithms, and improved system-level solutions for distributed deployment. We hope this survey serves as a valuable reference for researchers and practitioners working on MoE deployment, providing a structured understanding of current approaches and inspiring new directions in this

rapidly evolving field. To facilitate ongoing updates and the sharing of cutting-edge advances, we have established a repository at https://github.com/MoE-Inf/awesome-moe-inference/, which we plan to update semiannually to ensure its continued relevance as a dynamic resource for the community.

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